

ABELEJE OLANIYI GEORGE

Senior Backend Engineer · AI Systems & Real-Time Voice Platforms
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SUMMARY

Senior backend engineer with 5 years of production experience building scalable Python services, LLM-powered agent systems, and real-time distributed architectures using FastAPI, Django, PostgreSQL, Redis, GCP and AWS. I specialize in AI orchestration; structuring agents with tool-calling state machines, memory layers, and event-driven workflows that reason and act, not just respond.

I bring clean architecture discipline (DDD, abstract ports, domain isolation) without sacrificing shipping speed. My recent work spans real-time voice AI platforms and cooperative finance systems: both built for production reliability under live user load.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript Backend: FastAPI, Django (DRF), Node.js/Express, RESTful API Design, Webhooks

Frontend: React, Next.js, Tailwind CSS

AI/LLM: Gemini Live API, LLM tool-calling, agent state machines, RAG pipelines, prompt engineering

Databases: PostgreSQL, MongoDB, Redis, SQLite

Cloud & DevOps: AWS (EC2, S3, RDS, SES), Docker, CI/CD pipelines, Render, Vercel

Other: Domain-Driven Design, WebRTC, distributed systems, performance optimization, load testing, system design.

PROFESSIONAL EXPERIENCE

Full-Stack Developer · **Winnov8**

2025 – Present

- Designed and delivered an intelligent recommendation system end-to-end, scoped with stakeholders, architected the service layer, and iterated based on real production usage.
- Optimized a real-time notification and messaging system, increasing throughput by ~58% under live load through async processing and queue tuning.
- Contributed to architecture decisions, code reviews, and performance tuning as the platform scaled.

Full-Stack Developer · **Revela Africa**

2025

- Built core event-management backend services: ticketing, dynamic RSVP, secure payments, and authentication — production-ready and device-agnostic.

- Integrated Google APIs and payment gateways with webhooks for automated onboarding, calendar syncing, and checkout orchestration.
- Stabilized performance under peak traffic using Redis-backed service orchestration, significantly reducing response times.

Backend Engineer · **The ECO Platform**

2024

- Audited and refactored Django codebase — eliminated dead code, rewrote inefficient ORM query paths, and enforced clean architecture across service boundaries.
- Reduced API latency by over 91% and scaled throughput to support thousands of concurrent users through systematic query optimization and caching.

Backend Engineer · **WispTalk Africa**

2023 – 2024

- Led cloud migration to AWS (S3, RDS, EC2, CloudFront), significantly improving uptime and production resilience.
- Established monitoring practices and incremental performance improvements across live workloads.

Freelance Software Developer · **Independent**

2020 – 2022

- Built automation pipelines, data scrapers, and backend systems for clients across industries — independently managed scoping, development, and delivery.

PROJECTS

Truefit.ai – Real-time AI voice interview platform

Stack: Python, FastAPI, WebRTC (aiortc), Gemini Live API, PostgreSQL, Redis, Next.js, TypeScript

- Architected a dual-channel real-time system: WebSocket for session signalling and WebRTC (SRTP/UDP) for media — deliberately chosen over naive WebSocket streaming to eliminate head-of-line blocking and ensure jitter-resilient audio delivery.
- Structured Gemini as a tool-calling interview state machine with domain-driven architecture and abstract ports, giving the agent zero dependency on current infrastructure — clean algorithmic decision logic surfaced through a Next.js UI.
- Diagnosed and fixed a compound audio pipeline bug (greedy queue drain collapsing 40 chunks + pts clock reset corrupting frame pacing), resolving an infinite greeting loop that made the system unshippable.

CoopWise – AI-powered group savings & cooperative management platform

Stack: FastAPI, PostgreSQL, Redis, Next.js, Zustand, Gemini 1.5 Flash, Smart Contracts (Flow)

- Architected scalable FastAPI backend handling authentication, contribution cycles, payouts, and real-time financial state management with Redis-backed caching.
- Implemented algorithmic scoring and AI agents with RAG pipelines to surface personalised financial insights — decision logic built end-to-end from data model to React UI.
- Earned hackathon recognition for system design, scalability, and real-world impact.

EDUCATION

B.Sc. Computer Science · National Open University of Nigeria (*Final year, 2026*)

Professional Development, Backend Engineering · ALX Africa (*Mar 2025 – Oct 2025*)